

PMA Note

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July 14, 2026

1 The Real and Complex Number Systems

1.1 Introduction

Example 1.1. It is well known that there is no rational p such that $p^2 = 2$.

Define $A := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$ and $B := \{p \in \mathbb{Q}^+ \mid p^2 > 2\}$. Then A contains no largest number and B contains no smallest.

Proof. It suffices to show that

$$\forall p \in A, \exists q \in A \text{ s.t. } p < q$$

and

$$\forall p \in B, \exists q \in B \text{ s.t. } q < p.$$

For every $p \in \mathbb{Q}^+$, take

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \quad (1)$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \quad (2)$$

Hence,

$$\begin{aligned} p \in A &\implies p^2 - 2 < 0 \\ &\implies p < q \text{ and } q^2 < 2 && \because (1), (2) \\ &\implies q \in A \end{aligned}$$

and

$$\begin{aligned} p \in B &\implies p^2 - 2 > 0 \\ &\implies 0 < q < p \text{ and } q^2 > 2 && \because (1), (2) \\ &\implies q \in B. \end{aligned}$$

We have shown the desired result. $\square$

1.2 Ordered Sets

Definition 1.2. Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties:

- (i) If $x \in S$ and $y \in S$ then one and only one of the statements

$$x < y, \quad x = y, \quad y < x$$

is true.

(ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

Definition 1.3. An *ordered set* is a set S in which an order is defined.

Definition 1.4. Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E .

Lower bounds are defined in the same way (with $\geq$ in place of $\leq$).

Definition 1.5. Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

- (i) α is an upper bound of E .
- (ii) If $\gamma < \alpha$ then γ is not an upper bound of E .

Then α is called the *least upper bound* of E [that there is at most one such α is clear from (ii)] or the *supremum* of E , and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E .

Definition 1.6. An ordered set S is said to have the *least-upper-bound property* if the following is true:

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S .

Remark 1.7. The ordered set $\mathbb{Q}$ does not have the least-upper-bound property.

The following theorem shows that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

Theorem 1.8. Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B . Then

$$\alpha = \sup L$$

exists in S , and $\alpha = \inf B$.

In particular, $\inf B$ exists in S .

Proof. Since B is bounded below, L is nonempty. Let $x \in B$. Then $y \leq x$ for every $y \in L$. Thus, every $x \in B$ is an upper bound of L . Since L is a nonempty subset of S that is bounded above, by hypothesis,

$$\alpha = \sup L$$

exists in S .

If $\gamma < \alpha$ then γ is not an upper bound of L , hence $\gamma \notin B$. It follows that $\alpha \leq x$ for every $x \in B$. Thus, $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L .

We conclude that α is a lower bound of B (since $\alpha \in L$) and if $\alpha < \beta$ then β is not a lower bound of B (since $\beta \notin L$), which implies $\alpha = \inf B$. $\square$

Remark 1.9. Fix $x \in B$. Then $y \leq x$ for all $y \in L$. Thus, $\sup L \leq x$. Since the choice of $x \in B$ was arbitrary, we have

$$\sup L = \alpha \leq x \text{ for every } x \in B,$$

which implies α is a lower bound of B . Hence, $\alpha \in L$.

1.3 Fields

Definition 1.10. A *field* is a set F with two operations, called *addition* and *multiplication*, which satisfy the following so-called "field axioms" (A), (M), and (D):

(A) Axioms for addition

- (A1) If $x \in F$ and $y \in F$, then their sum $x + y$ is in F .
- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in F$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- (A4) F contains an element 0 such that $0 + x = x$ for every $x \in F$.
- (A5) To every $x \in F$ corresponds an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If $x \in F$ and $y \in F$, then their product xy is in F .
- (M2) Multiplication is commutative: $xy = yx$ for all $x, y \in F$.
- (M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in F$.
- (M4) F contains an element $1 \neq 0$ such that $1x = x$ for every $x \in F$.
- (M5) If $x \in F$ and $x \neq 0$ then there exists an element $1/x \in F$ such that

$$x \cdot (1/x) = 1.$$

(D) The distributive law

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in F$.

Definition 1.11. An *ordered field* is a field F which is also an *ordered set*, such that

- (i) $x + y < x + z$ if $x, y, z \in F$ and $y < z$.
- (ii) $xy > 0$ if $x \in F$, $y \in F$, $x > 0$, and $y > 0$.

1.4 The Real Field

Theorem 1.12. *There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.*

Theorem 1.13.

(a) *If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that*

$$nx > y.$$

(b) *If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.*

Proof.

(a) Let $x, y \in \mathbb{R}$ and $x > 0$. Suppose to the contrary that $nx \leq y$ for each $n \in \mathbb{N}$ and define

$$A := \{nx \mid n \in \mathbb{N}\}.$$

Then A is nonempty and bounded above by y . Thus, the supremum of A exists in $\mathbb{R}$. Put $\alpha = \sup A$.

Since $x > 0$, we have $\alpha - x < \alpha$, hence, $\alpha - x$ is not an upper bound of A . Therefore, $\alpha - x < mx$ for some $m \in \mathbb{N}$. Then $\alpha < (m + 1)x \in A$, which is a contradiction, since α is an upper bound of A .

(b) Since Rudin's proof of part (b) is already explicit, we omit it here. $\square$

Theorem 1.14. *For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$.*

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof. We omit the proof here. $\square$

Corollary 1.15. *If a and b are positive real numbers and n is a positive integer, then*

$$(ab)^{\frac{1}{n}} = a^{\frac{1}{n}} b^{\frac{1}{n}}.$$

Proof. We omit the proof here. $\square$

Lemma 1.16.

(a) *If $E \subset \mathbb{Z}$ has the infimum, then $\inf E \in E$.*

(b) *If $x \in \mathbb{R}$ then there exists an $n \in \mathbb{Z}$ such that*

$$n \leq x < n + 1.$$

Proof.

(a) Let $\alpha = \inf E$. Suppose to the contrary that $\alpha \notin E$. Since $\alpha + 1$ is not a lower bound of E , choose an $x_0 \in E$ such that

$$\alpha < x_0 < \alpha + 1. \quad (3)$$

Since x_0 is not a lower bound of E , choose an $x_1 \in E$ again such that

$$\alpha < x_1 < x_0. \quad (4)$$

Subtracting x_0 from (4), we obtain $\alpha - x_0 < x_1 - x_0 < 0$. Since $-1 < \alpha - x_0$ from (3), it follows that $-1 < x_1 - x_0 < 0$. This contradicts the fact that $x_1 - x_0 \in \mathbb{Z}$. We conclude that $\alpha \in E$.

(b) Let $x \in \mathbb{R}$ and define $E := \{n \in \mathbb{Z} \mid n > x\}$. By the Archimedean Property, E is nonempty. Since E is bounded below by x , the infimum of E exists in $\mathbb{R}$. Since $E \subset \mathbb{Z}$, $\inf E$ is the least element of E by (a). Thus, we have

$$\inf E - 1 \leq x < \inf E.$$

Putting $n = \inf E - 1 \in \mathbb{Z}$, we obtain the desired inequality. $\square$

Remark 1.17 (Decimals). Let $x > 0$ be real. Let n_0 be the largest integer such that $n_0 \leq x$, by lemma 1.16. Having chosen $n_0, n_1, \dots, n_{k-1}$, let n_k be the largest integer such that

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq x.$$

Let E be the set of these numbers

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \quad (k = 0, 1, 2, \dots). \quad (5)$$

Notice that x is an upper bound of E and if $k \in \mathbb{N}$ then

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq x < n_0 + \frac{n_1}{10} + \dots + \frac{n_k + 1}{10^k}.$$

Hence, we have

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < \frac{1}{10^k} \text{ for } k \in \mathbb{N}. \quad (6)$$

Suppose $y < x$. Then by the Archimedean Property, there exists a positive integer k such that $1/10^k < x - y$. From (6), we obtain

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < x - y,$$

hence

$$y < n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k}.$$

Therefore, $x = \sup E$. The decimal expansion of x is

$$n_0 \cdot n_1 n_2 n_3 \cdots. \quad (7)$$

Conversely, for any infinite decimal (7), the set E of numbers (5) is bounded above, and (7) is the decimal expansion of $\sup E$.

Exercises

Exercise 1. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A).$$

Proof. Since A is bounded below, the infimum of A exists in $\mathbb{R}$. If $a \in A$ then $\inf A \leq a$. Hence,

$$-a \leq -\inf A \text{ for all } a \in A, \quad (8)$$

and it follows that $-A$ is bounded above. Thus, $\sup(-A)$ exists in $\mathbb{R}$. Since $-\inf A$ is an upper bound of $-A$ from (8), we have $\sup(-A) \leq -\inf A$, hence

$$\inf A \leq -\sup(-A). \quad (9)$$

Since $-a \leq \sup(-A)$ for every $a \in A$, we obtain $-\sup(-A) \leq a$ for all $a \in A$. Therefore, $-\sup(-A)$ is a lower bound of A and

$$-\sup(-A) \leq \inf A. \quad (10)$$

Combining (9) and (10), we conclude that $\inf A = -\sup(-A)$. $\square$

Exercise 2. Fix $b > 1$.

(a) If m, n, p, q are integers, $n > 0, q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

(c) If x is real, define $B(x)$ to be the set of all numbers b^t , where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x .

(d) Prove that $b^{x+y} = b^x b^y$ for all real x and y .

Proof.

(a) Let $b^{mq} = b^{np} = x$. Since n and q are positive integers and $b > 0$, by Theorem 1.14, we have

$$b^m = x^{1/q} \text{ and } b^p = x^{1/n}.$$

It follows that

$$(b^m)^{1/n} = (x^{1/q})^{1/n} = (x^{1/n})^{1/q} = (b^p)^{1/q},$$

since $[(x^{1/q})^{1/n}]^{nq} = [(x^{1/n})^{1/q}]^{nq} = x$.

(b) Let m, n, p, q be integers and $n > 0, q > 0$ and let $r = m/n$ and $s = p/q$. Then by Corollary 1.15, we obtain

$$\begin{aligned} b^{r+s} &= b^{\frac{m}{n} + \frac{p}{q}} = b^{\frac{mq+np}{nq}} \\ &= (b^{mq+np})^{\frac{1}{nq}} \\ &= (b^{mq} b^{np})^{\frac{1}{nq}} \\ &= (b^{mq})^{\frac{1}{nq}} (b^{np})^{\frac{1}{nq}} \\ &= b^{\frac{mq}{nq}} b^{\frac{np}{nq}} = b^{\frac{m}{n}} b^{\frac{p}{q}} = b^r b^s. \end{aligned}$$

(c) Note that by Theorem 1.14, $b^r > 0$ if $b > 1$ and r is rational.

Let n be a positive integer and let $0 < a < 1$ be real. We first show that $0 < a^{1/n} < 1$.

Suppose to the contrary that $a^{1/n} \geq 1$. Then we obtain

$$1 \leq \left(a^{1/n}\right)^n = a < 1,$$

a contradiction. Therefore, if $0 < a < 1$ then $0 < a^{1/n} < 1$, where n is a positive integer.

Suppose r and s are rational. Let m, n, p , and q be integers and let n and q be positive. Then we write $r = m/n, s = p/q$.

Suppose $r < s$. Then

$$r - s = \frac{m}{n} - \frac{p}{q} = \frac{mq - np}{nq} < 0,$$

hence

$$0 < b^{mq-np} = \frac{1}{b^{np-mq}} < 1$$

and

$$0 < b^{r-s} = b^{\frac{mq-np}{nq}} = (b^{mq-np})^{1/nq} < 1.$$

Thus, if $r < s$ then

$$b^r = b^{(r-s)+s} = b^{r-s} b^s < b^s.$$

If t is rational and $t \leq r$ then $b^t \leq b^r$. It follows that $B(r)$ is bounded above by b^r and $\sup B(r)$ exists in $\mathbb{R}$. Since b^r is an upper bound of $B(r)$,

$$\sup B(r) \leq b^r.$$

Since $b^r \in B(r)$,

$$b^r \leq \sup B(r).$$

We conclude that $b^r = \sup B(r)$.

(d) Let t_1 and t_2 be rational. If $t_1 \leq x$ and $t_2 \leq y$ then $t_1 + t_2 \leq x + y$. Thus, we have

$$b^{t_1}b^{t_2} = b^{t_1+t_2} \leq \sup B(x+y) = b^{x+y}.$$

Fix $t_1 \leq x$. Then

$$b^{t_2} \leq b^{x+y}/b^{t_1} \text{ for all } t_2 \leq y.$$

Hence,

$$b^y = \sup B(y) \leq b^{x+y}/b^{t_1}.$$

Since the choice of t_1 was arbitrary,

$$b^{t_1} \leq b^{x+y}/b^y \text{ for all } t_1 \leq x,$$

and it follows that

$$b^x = \sup B(x) \leq b^{x+y}/b^y.$$

Therefore, we obtain

$$b^x b^y \leq b^{x+y}. \quad (11)$$

Notice that

$$0 < b^{t_1} \leq b^x \text{ for every } t_1 \leq x$$

and

$$0 < b^{t_2} \leq b^y \text{ for every } t_2 \leq y.$$

Hence, for every $t_1 \leq x$ and $t_2 \leq y$,

$$b^{t_1+t_2} = b^{t_1}b^{t_2} \leq b^x b^y.$$

Now, to obtain

$$b^{x+y} = \sup B(x+y) \leq b^x b^y \quad (12)$$

from above inequality, we need the following statement:

$$b^x = \sup \{b^t \mid t \in \mathbb{Q}, t < x\}.$$

Let us prove it. Notice that

$$b^n - 1 = (b-1)(b^{n-1} + \cdots + 1) \geq n(b-1)$$

for any positive integer n . Put $c = b^{1/n} > 1$. Then we have

$$b-1 \geq n(b^{1/n} - 1).$$

Hence, if $t > 1$ and $n > (b-1)/(t-1)$ then

$$\frac{n(b^{1/n} - 1)}{t-1} \leq \frac{b-1}{t-1} < n,$$

and it follows that

$$b^{1/n} < t.$$

Note that if $t > 1$ then $b^{1/n} < t$ for sufficiently large n .

Let $B_0(x) = \{b^t \mid t \in \mathbb{Q}, t < x\}$. We need to show that b^x is the supremum of $B_0(x)$. It is clear that b^x is an upper bound of $B_0(x)$. Thus, it remains to prove that if $y < b^x$ then y is not an upper bound.

Suppose $0 < y < b^x$. Then $b^x/y > 1$ and it follows that

$$b^{1/n} < b^x/y$$

for sufficiently large n . Hence, we have

$$y < \frac{b^x}{b^{1/n}} < b^x.$$

By the density of rationals, choose a rational t such that $x - (1/n) < t < x$. Since $x < s < t + (1/n)$ for some rational s ,

$$b^x \leq b^s < b^{t+(1/n)} = b^t b^{1/n}$$

and

$$y < \frac{b^x}{b^{1/n}} < b^t.$$

Thus, y is not an upper bound of $B_0(x)$ and therefore $b^x = \sup B_0(x)$.

Now, let t be a rational and let $t < x + y$. Pick $t_2 \in \mathbb{Q}$ such that $t - x < t_2 < y$ and set $t_1 = t - t_2$. Then for every $t \in \mathbb{Q}$ with $t < x + y$, there are rational $t_1 < x$ and $t_2 < y$ such that $t = t_1 + t_2 < x + y$, hence $b^t = b^{t_1+t_2} = b^{t_1} b^{t_2} \leq b^x b^y$. Thus, we obtain (12).

Combining (11) and (12), we conclude that

$$b^{x+y} = b^x b^y.$$

□

Exercise 3. Prove that no order can be defined in the complex field that turns it into an ordered field.

Proof. Suppose to the contrary that there exists an order $<$ on $\mathbb{C}$ which turns $\mathbb{C}$ into an ordered field. Let $\mathbf{0}$ and $\mathbf{1}$ denote the additive and multiplicative identities of the complex field, respectively. Then $\mathbf{0} = (0, 0)$ and $\mathbf{1} = (1, 0)$.

Note that $i = (0, 1) \neq \mathbf{0}$. If $x \neq \mathbf{0}$ then $x^2 > \mathbf{0}$. It follows that

$$i^2 = (0, 1)(0, 1) = (-1, 0) > \mathbf{0}.$$

Since $(-1, 0) + (1, 0) = \mathbf{0}$, $(-1, 0)$ is the additive inverse of $(1, 0)$. Since $\mathbf{1} > \mathbf{0}$ and if $x > \mathbf{0}$ then $-x < \mathbf{0}$ in any ordered field, we obtain

$$(-1, 0) = -(1, 0) = -\mathbf{1} < \mathbf{0}.$$

This contradicts the Trichotomy Property.

□

2 Basic Topology

2.1 Finite, Countable, and Uncountable Sets

We omit this section here.

2.2 Metric Spaces

Definition 2.1. A set X , whose elements we shall call *points*, is said to be a *metric space* if with any two points p and q of X there is associated a real number $d(p, q)$, called the *distance* from p to q , such that

- (a) $d(p, q) > 0$ if $p \neq q$; $d(p, p) = 0$;
- (b) $d(p, q) = d(q, p)$;
- (c) $d(p, q) \leq d(p, r) + d(r, q)$, for any $r \in X$.

Any function with these three properties is called a *distance function*, or a *metric*.

Example 2.2. $\mathbb{R}^k$ is a metric space.